Research proof for review

Optimal fractional vertex cover under vertex arrivals

Revised November 2, 2026

In the vertex-arrival model, each new vertex reveals its edges to earlier vertices. A fractional algorithm maintains a value $y_v \in [0,1]$ on each vertex. Values can only increase, and every revealed edge uv must satisfy $y_u + y_v \geq 1$. On an unweighted input, ALG is the final sum of these values and OPT_f is the minimum cost of an offline fractional cover.

Theorem. For every $\epsilon > 0$ and every randomized online fractional vertex-cover algorithm, there is a finite, unweighted bipartite vertex-arrival input, independent of the algorithm's realized random bits, with $\mathrm{OPT}_f > 0$ and

$$\mathbb{E}[\mathrm{ALG}] > (B^* - \epsilon)\,\mathrm{OPT}_f.$$

Here $k > 1$ solves $k = \coth k$, $a = (k+1)/2$, $b = (k-1)/2$, and

$$B^* = 1 + a^{a/k} b^{b/k} = 1.9007616968738465\ldots.$$

The core inputs are blowups of rooted trees. Each new pool is adjacent to just one old pool, and old pools can receive arbitrarily many children. Together with Wang and Wong's upper bound [1], this gives the optimal ratio also for the general-graph fractional problem.

The proof keeps track of the worst cost separately according to whether the offline cover includes the root. After a child arrives there are three compatible pairs of root choices. A full-history strategy might try to spend less by observing which of the two branches grows first. Sections 2 and 3 show that the adversary can order the two tests so this information has no value at either branch's decisions. Section 4 converts those conditional costs into a feasible function for Wang and Wong's minimax problem.

1 A countable game and its conditional costs

Normalize the root pool to weight one. A core move adds an independent pool of announced positive rational weight joined completely to its named parent. There are no other new edges. Number pools in local arrival order and use the opposite color of the parent. Allow every finite such history, including a stop just after the root. An online response gives monotone coordinates in $[0,1]$ satisfying each edge inequality. A pool represents an average over twins. Section 5 removes the batch and rational-weight conventions.

Fix a rational $0 < \beta < B^* - 1$. For any integral cover C of a core prefix, put

$$L(C) = \sum_{v \notin C} w_v y_v - \beta \sum_{v \in C} w_v.$$

To see how a positive surplus forces a violation, use a common integer scale for all pool weights in this prefix. Attach a private leaf to each individual vertex represented by C. The completed graph has an offline cover C and a disjoint private matching of the same

cardinality. Monotonicity preserves the weight already paid outside C. Each private matching edge requires at least one unit from its endpoints. Measuring costs in units of the common scale, the algorithm pays at least $w(C) + \sum_{v \notin C} w_v y_v$, and $\mathrm{OPT}_f = w(C)$, where $w(C) = \sum_{v \in C} w_v$. Lu uses a complete-bipartite extension with the same matching accounting [2]. The edge-arrival construction of Tang and Zhang uses private edges [3].

Start the isolated root at zero. Assume for contradiction that every finite family of core prefixes admits a strategy with $L(C) \le 0$ for every prefix and cover in the family. For real $p \ge 0$, define $F(p)$ as the infimum of the values q for which a **single causal strategy on all core histories** can maintain both promises

$$L(C) \le p \quad \text{(root outside } C\text{)}, \qquad L(C) \le q - \beta \quad \text{(root inside } C\text{)}. \tag{1}$$

The strategy space uses countably many coordinates in $[0, 1]$, one vector per public history. Feasibility, monotonicity and the promises at a given prefix are closed finite-coordinate conditions. Compactness, equivalently diagonal subsequence extraction, gives one strategy for all histories. Thus $F(0) \le \beta$. The root-only inside test gives $F \ge 0$. The infimum is attained by compactness. The set of attainable pairs (p, q) is the epigraph of F. It is closed by compactness of the strategy space and convex by coordinatewise mixing of strategies. Consequently $0 \le F \le \beta$, F is nonincreasing and convex, and it is continuous on $[0, \infty)$, including at zero by closedness and monotonicity. No scalar state assumption has been made about the algorithm.

2 Interleaving two finite tests

A *local test* is a pair consisting of a finite rooted input and a terminal cover of that input. Only the sequence of core arrivals is public. The terminal cover is a hidden label. A finite distribution of tests is a tree of public prefixes, with termination and its hidden label as possible final outcomes. Several outcomes may have the same last public prefix. Both roots and their connecting edge are already present. A local arrival in the lemma is a subsequent arrival in that component.

Interleaving lemma. Take two finite rational distributions of tests on disjoint rooted components A, B. Write $\alpha = \Pr(A\text{'s root outside})$ and $\delta = \Pr(B\text{'s root inside})$. If $\alpha \le \delta$, there is a distribution of interleaved tests with these properties.

1. Both local marginals are preserved, including their entire input and hidden cover. The two selected covers jointly cover the connecting root edge on every outcome of positive probability.

2. At **each local arrival** in either component, conditional on its local public prefix, the joint public past available for that decision gives no further information about the rest of that component's test. This includes its hidden cover.

Construction. In the first local prefix tree mark each node by the conditional probability that its terminal cover excludes its root. In the second tree use the probability that its cover includes its root. Terminal labels are marked zero or one. Each mark is the probability-weighted mean of its children's marks. Implement a multiway transition by a finite binary tree of choices, marking each internal split with its corresponding weighted mean. Only the

resulting next public arrival is revealed. Zero-probability outcomes can be dropped. Choose a common denominator M for all marks.

Place two particles at $M\alpha, M\delta$ in $\{0, \dots, M\}$. They use the **same sequence** of independent fair signs. At each tick add the sign to each interior particle and leave an absorbed particle fixed. For a local binary choice with distinct child marks, wait until its particle first hits either scaled child mark on this lattice and take that branch. From m between child positions $l < u$, the probability of hitting u first is $(m-l)/(u-l)$. This equals the prescribed mass of the upper child since the parent mark is the mean of the two child marks. For equal child marks use a fresh independent coin with the prescribed probability and advance no ticks. Continue this procedure through local arrivals and the final hidden label. Once absorbed, all remaining choices in that local tree have equal marks and use the coin rule.

Merge the local arrivals by tick, keeping local order. Break ties by a fixed component priority. Reveal only actual arriving pools and their parents. Hidden transitions and termination labels are not announcements. The particles remain ordered at every tick, including when one is absorbed. Every required hit occurs almost surely and there are only finitely many local transitions. At termination the two binary marks are ordered. Thus if A's root is outside, B's root is inside. Although the number of private walk ticks is unbounded, the public input ranges over only finitely many local paths and shuffles.

Fix a local public prefix v in component A. Write T_A for its full test and $v \preceq T_A$ for the event that the test reaches v. On this event let H_v be the joint public history up to and including the arrival at v. For every possible H ending there and every full local test τ, we claim

$$\Pr(T_A = \tau \mid H_v = H) = \Pr(T_A = \tau \mid v \preceq T_A).$$

Fix the other component's independent coins. On reaching v, record the finite word of signs used and the outcomes of the own equal-mark choices made through that arrival. End the record before the next own choice, even if it will use the same tick. For any particular record, the event of reaching v with that record depends only on the signs and own choices it contains. The unused signs and fresh own coins therefore have their original law. The sampler resumes at the node for v, at that node's mark, so the remaining test has precisely the standalone law at v.

Given the other coins, this record also determines H_v. If the record uses m signs, it determines all other arrivals before tick m. Fixed component priority specifies whether the other arrivals at tick m precede this decision. Those arrivals use only the first m signs and the other coins. The conditional law is the same for each such record, so averaging over records and other coins consistent with H proves the claim. The same argument applies on B.

3 Why the budgets must separate

Fix $p \geq 0$ and a strategy attaining (1) at $q = F(p)$. Offer a first child of rational weight $h > 0$. Fix the resulting root coordinates r, z, with $r + z \geq 1$. There are now two rooted components if their connecting edge is set aside. Each can undergo every rational rooted-tree continuation, in any interleaving. Scale the child's subsequent weights by h. Its local surplus will be expressed in units of h. Arrival indices in each local test can be translated canonically to global indices.

We claim there exist **two independent local strategies**, starting at r, z, that meet the global promises against *every compatible pair* of local tests. This is a consequence of the interleaving lemma, even though the original strategy sees both histories.

For each side, a local strategy assigns a response to each of its own prefixes. Independence here means that each assignment uses only that side's history. If there were no such pair, compactness of the product of the two strategy spaces would give a finite list of test pairs whose inequalities no pair of local strategies satisfies. For each test pair the inequality is that $L_A + hL_B$ be at most p or $q - \beta$, according to A's root status. Compatibility here means the two roots are not both outside. The product of local strategy polytopes on these finite prefix trees is compact and convex, and each performance inequality is affine. Write $e_i(x)$ for the excess above the threshold on test pair i, where x ranges over that product polytope. Then $\min_x \max_i e_i(x) > 0$, since the minimum is attained. By finite minimax there are weights $\lambda_i \geq 0$ summing to one for which $\min_x \sum_i \lambda_i e_i(x) > 0$. Approximation within the probability simplex preserves this strict inequality because the finitely many e_i are bounded. We may therefore take rational weights supported on the same compatible test pairs.

Keep only the two local marginals of this distribution. Their ordered root probabilities allow the interleaving construction in Section 2. The expected threshold depends only on A's marginal root status. The expected performance of independent strategies likewise depends only on the two marginals.

To apply separation to the joint strategy, simulate it internally from the fixed coordinates r, z. At each subsequent arrival copy its newborn coordinate y. Replace each older neighbor's current value x by $\max(x, 1 - y)$, and keep every other old value. The simulated neighbor value is at least x by induction and at least $1 - y$ by feasibility. Thus each coordinate stays at most its simulated value, and all old and new edges are covered. This strategy meets the original promises and changes each component only at its own arrivals.

Let Y_v be its vector on A just after local arrival v. This vector is a function of H_v. Define the local strategy by $\bar{y}_v = \mathbb{E}[Y_v \mid v \preceq T_A]$. The lemma gives

$$\mathbb{E}[Y_v \mid T_A = \tau] = \bar{y}_v \qquad (v \preceq \tau)$$

for every full test of positive mass, including its hidden cover. For a child prefix v' of v, average this equality over the tests reaching v'. On coordinates already present at v this gives

$$\bar{y}_v = \mathbb{E}[Y_v \mid v' \preceq T_A] \;\leq\; \mathbb{E}[Y_{v'} \mid v' \preceq T_A] = \bar{y}_{v'}.$$

The local edge inequalities hold by linearity. Start at r and set all coordinates to one on prefixes outside the sampled tree. Do the same on B, starting at z.

Given a full local test, the final local vector equals the vector at its last local arrival, or the fixed initial vector if there was none. Thus the expected surplus conditional on this test equals the surplus of its local mean strategy. Summing over the two marginals, the joint strategy has the same expected excess as this pair of local strategies. Separation makes that excess strictly positive, contradicting the promises on every interleaving.

For the independent strategies let s, t be the suprema of their outside-root surpluses. Pairing each with the other root's root-only inside test gives $s \leq p + h\beta$ and $ht \leq q$, so both are finite. They are at least r, z by the root-only outside tests. A local strategy starting at a positive coordinate can be simulated from zero by postponing that initial purchase until the first local move. This can only improve its root-only outside test. By the definition of F, the

supremal inside surpluses are at least $F(s) - \beta$ on the parent side and $F(t) - \beta$ on the child side. The two inputs can be chosen separately for any compatible pair of root statuses. Taking suprema in the outside/inside and inside/outside inequalities shows that for every $p \geq 0$ and rational $h > 0$ there exist $s, t \geq 0$ satisfying

$$s + h(F(t) - \beta) \leq p, \qquad F(s) + ht \leq F(p), \qquad s + t \geq 1. \tag{2}$$

The initial promises p need not be reachable from a particular play. Their frontier was defined using the very same rooted input family. At the isolated root we obtain $F(0) \leq \beta$.

3.1 A fork that can be checked directly

As a check on the three compatible root choices, take two adjacent unit pools and allow one further child of weight $h = 1/2$ on each endpoint, in either order. The exact excess ratio with terminal completions is $3/5$. Label the endpoints U, V so that V's initial coordinate is at least $1/2$. Attach the first child to U, and let r be U's coordinate then. The cover consisting of U requires $1/2 + h(1 - r) \leq \beta$. Attach the second child to V and write s for V's coordinate. The two covers that exclude one endpoint require

$$r + h(1 - s) \leq \beta(1 + h), \qquad s + h(1 - r) \leq \beta(1 + h).$$

At $h = 1/2$, weight the one-child inequality by 3 and the two displayed inequalities by 2 and 1, respectively. The coefficients of r, s cancel, giving $9/2 \leq (15/2)\beta$, hence $\beta \geq 3/5$. Conversely keep the isolated root at zero until the connecting edge arrives, then set both endpoints to $1/2$. If a child arrives, raise its parent to $4/5$ and give the child $1/5$. At $\beta = 3/5$ the outside surplus at an endpoint with a child is $4/5 - (3/5)(1/2) = 1/2$. Its inside surplus is at most $(1/2)(1/5) - 3/5 = -1/2$. Without a child the corresponding bounds are $1/2$ and $-3/5$. Every compatible pair therefore has nonpositive total surplus. On completion give each private leaf the complement of its neighbor's current value. Each matched pair now costs exactly one, so the completed cost is at most $(1+\beta)w(C)$. In the blowup by two, a core arrival sees both twins of its parent, whereas a private leaf sees a single vertex. Thus the completion rule can be implemented from the revealed edges alone.

4 Recovering the exact constant

Put $D = \beta - F$. It is nonnegative, concave, and bounded by β. For a differentiability point $0 < p < 1$, convexity and (2) give

$$t \leq -F'(p)D(t), \qquad t \geq 1 - p - h\beta.$$

Indeed $ht \leq -F'(p)(s - p)$ by the supporting-line inequality, and $s - p \leq hD(t)$ with $-F'(p) \geq 0$. For $\eta > 0$ write $g_\eta(v) = v/(D(v) + \eta)$. Concavity and $D(0) \geq 0$ give $D(x) + \eta > (x/y)(D(y) + \eta)$ for $0 \leq x < y$, so g_η is strictly increasing. For every sufficiently small rational $h > 0$,

$$-F'(p) \;\geq\; g_\eta(t) \;\geq\; g_\eta(1 - p - h\beta).$$

The right side converges as $h \downarrow 0$, regardless of the choice of t for each h. Hence

$$-F'(p) \;\geq\; \frac{1-p}{\beta+\eta-F(1-p)}.$$

Define $f_\eta(u) = \beta + \eta - F(1-u) > 0$ on the closed unit interval. Integrate from p to 1, using local absolute continuity of convex functions, $F(1) \geq 0$, and continuity to extend to $p = 0$. Then

$$f_\eta(1-p) + \int_p^1 \frac{1-u}{f_\eta(u)}\,du \;\leq\; \beta+\eta \qquad (0 \leq p \leq 1). \tag{3}$$

The function f_η is continuous, positive, and $(1-u)/f_\eta(u) = g_\eta(1-u)$ is decreasing. Wang and Wong [1, Section 4.1, Eq. (1)] minimize the supremum of the left side of (3) plus one over this class. Their equalization argument and solution of the differential equation reduce the optimum excess to minimizing $G(k) = a^{a/k}b^{b/k}$ with $a = (k+1)/2$, $b = (k-1)/2$. For $k > 1$,

$$\frac{d}{dk}\log G(k) = \frac{k - \frac{1}{2}\log((k+1)/(k-1))}{k^2}.$$

The numerator increases from $-\infty$ to $+\infty$ and vanishes at $k = \coth k$. This gives the minimum, including comparison with the endpoint limit $G(1) = 1$. Thus (3) forces $B^* - 1 \leq \beta + \eta$ for every $\eta > 0$, contradicting $\beta < B^* - 1$.

5 Constructing a finite ordinary input

For a requested ϵ, choose a rational $\max(0, B^*-1-\epsilon) < \beta < B^*-1$. The following terminating search specifies finite adversarial families even when a manageable size bound is unavailable.

1. For $n = 1, 2, \ldots$, take all core histories with at most n pools, root weight one, nonroot weights a/b with $1 \leq a, b \leq n$, and every single-parent choice. Prefixes are shared across all continuations.

2. Test rational linear feasibility with one vector $y \in [0,1]^k$ per public prefix of k pools. Impose monotonicity from its parent prefix, each new edge inequality, zero at the isolated root, and $L(C) \leq 0$ for every integral pool cover at that prefix. Exact rational linear programming or Fourier–Motzkin elimination decides each stage. Stop at the first infeasible stage. If every stage were feasible, compactness would give the strategy ruled out by Sections 1–4. The weights used in the scaled child subgames are rational and thus all occur in this exhaustion.

3. Multiply all weights in this stage by $\operatorname{lcm}(1, \ldots, n)$. Replace each pool by that many independent twin vertices, presented consecutively, with all edges to the already present parent pool. If the horizon is announced, fix it now at the maximum length of all completed inputs of this stage, and pad shorter ones at the end with isolated vertices. On each fixed expanded prefix take expectations over the algorithm's bits and average over the twins of each pool. Averaging the edge inequalities over all pairs of twins across a parent-child edge gives the pool inequality. Use the same seed on a prefix and its

extensions to obtain monotonicity of these means. Set the vector at the root-only prefix to zero and keep the actual means at every later prefix. This preserves feasibility and monotonicity even for algorithms that buy at an isolated root.

4. Infeasibility supplies a prefix and cover C with strictly positive surplus on this evaluation. Finish that prefix with the private matching from Section 1. Both root-only tests have nonpositive surplus, so the prefix contains an edge and $w(C) > 0$. At this prefix the vector equals the algorithm's actual mean. Fix the prefix and its completion before sampling the algorithm's seed. In integer units the private matching and monotonicity outside C give

$$\mathbb{E}[\mathrm{ALG}] \;\geq\; w(C) + \sum_{v \notin C} w_v y_v \;>\; (1+\beta) w(C) \;>\; (B^* - \epsilon)\mathrm{OPT}_f.$$

All core and terminal edges are revealed with a new endpoint. The continuum limit in Section 4 tests rational offers of arbitrarily small size under the contrary hypothesis. The output graph itself uses no limiting weights or rounded player responses. The finite search is an existence construction and has no practical near-target runtime guarantee.

The script `scripts/tree_witness.py` implements the construction for a given algorithm. Its function `find_witness(beta, marginals)` evaluates `marginals(core_prefix)` on each scaled family, checks the cover budgets, and returns the completed unweighted arrival list, an offline cover, its optimum and the guaranteed ratio. This permits skipping the LP search for a family hard for every strategy. Given exact comparisons of marginal surpluses and a subcritical target, the search succeeds no later than the first infeasible stage.

6 A concrete finite certificate at 1.9005

The saved certificate in `experiments/fork-1801-2000.npz` describes a particular finite family without enumerating all its interleavings. The following recursion explains how to expand it and why it is a lower certificate. It also checks the direction of the inequalities in (2) without numerical differentiation or evaluating the Wang–Wong function.

A record with rational $a, b \geq 0$ denotes a finite rooted test family on which the promises (1) require $q + bp \geq a$. A root-only record with intercept zero is valid for every nonnegative slope. If a strategy starts the root with a positive coordinate, the same lower bound holds by simulation from zero. At a fork, join two records with lines $a_j - b_j s$ on the parent side and $a_i - b_i t$ on the child side. Offer the child of weight h, then allow all order-preserving shuffles of their core arrivals. A local stop leaves that component idle. Take private completions only after the global core stops, using a compatible pair of terminal covers. Include root-only stops in each component. The finite separation argument of Section 3 applies directly to these two finite test families at any feasible (p, q) because it uses only the global promises on their shuffles. For the resulting local strategies, let s, t be their maximum outside surpluses. The parent record bounds its maximum inside surplus below by $a_j - b_j s - \beta$, and the child record bounds its own by $a_i - b_i t - \beta$. The root-only outside tests give $s + t \geq 1$. Taking maxima for the two mixed root choices gives

$$p \geq s + h(a_i - \beta - b_i t), \qquad q \geq a_j - b_j s + ht.$$

For a desired $b > b_j$ with $bb_i < 1$, choose $h = (b - b_j)/(1 - bb_i) > 0$. Multiply the first inequality by b and add the second. The coefficients of s, t are now both $b - b_j$. Thus the forked family satisfies

$$q + bp \;\geq\; a_j + (b - b_j) - bh(\beta - a_i). \tag{4}$$

Rounding the right side down preserves validity. An alternative operation increases the slope of any old line at unchanged intercept using $p \geq 0$. This uses the same family.

Take $M = 4000,\ S = 4 \cdot 10^{12},\ \beta = 1801/2000$. Slopes are ℓ/M and intercepts are integer multiples of $1/S$. Row zero uses only root-only records. Record (k, ℓ) uses parent $(k, \ell - 1)$. A stored child index $i \geq 0$ uses child $(k - 1, i)$ and $h = M/(M^2 - \ell i)$ in (4). A value of -1 just increases the slope. Column zero is always the trivial line. These instructions are acyclic. They specify actual relative pool weights, allowed interleavings, and the arrival graphs for the witness in compressed form. Subtree weights are multiplied by the child's weight at each use and can all be given a common integer denominator on expansion.

At record $(126, 4105)$ the verified intercept numerator is 3,602,076,706,847, strictly above $S\beta = 3{,}602{,}000{,}000{,}000$. Thus $p = 0, q = \beta$ is impossible on that single family. The verifier recomputes the integer inequalities from the recorded child choices using Python arbitrary-precision integers. The certificate checks the finite claim conditional on the interleaving lemma. Sections 1–5 prove the limit independently of this example.

From the repository root, run the following command to replay the certificate.

```
python scripts/fork_certificate.py --check experiments/fork-1801-2000.npz
```

Running `python scripts/fork_certificate.py` without arguments regenerates it. The script `scripts/interleaving_check.py` separately checks a finite ordered walk with exact fractions, including coincident arrival ticks and an equal-mark split. A second fixture includes a hidden stop at the root, private splits, and distinct terminal labels with the same root status. It checks both tie priorities. In each case the verifier checks the full local test law at every decision prefix and zero mass on incompatible terminal labels. These finite checks supplement the proof of the general lemma.

References

[1] Yajun Wang and Sam Chiu-wai Wong. *Online Vertex Cover and Matching: Beating the Greedy Algorithm*. arXiv:1305.1694 (2013).

[2] Tianhang Lu. *A New Lower Bound for Online Vertex Cover under Vertex Arrivals*. arXiv:2608.09210 (2026).

[3] Zhihao Gavin Tang and Yuhao Zhang. *A Tight Bound on Online Vertex Cover under Edge Arrivals*. arXiv:2608.04994 (2026).